

36

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

COS:

Measure network performance using key metrics with simulation tools
(B1, B2, B3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks: 30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRIC FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	30%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints of problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

9

10

9

8

**COMPUTER NETWORKS
CONSTRAINT-BASED PROBLEM
SOLVING REPORT**

Course Code: CSA07
Course Outcome Covered: COS
Assessment Tool 1
(Expanded Edition)

Complete Question & Answer Document

Q2: A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Answer:

Designing a campus-wide wireless network for over 3,000 students with a limited number of Access Points (APs) requires highly strategic capacity planning and intelligent hardware selection to maximize coverage. The physical placement of the APs must follow a staggered, high-density deployment model. In high-traffic zones like libraries and massive lecture halls, Wi-Fi 6 (802.11ax) APs with directional antennas should be deployed to focus signals exactly where students gather. Wi-Fi 6 is specifically designed for high-density environments, handling multiple simultaneous connections much more efficiently through OFDMA technology. In hostels and corridors, omnidirectional APs should be placed in central hallways to maximize the coverage footprint with fewer physical devices.

To minimize signal interference, the network must primarily utilize the 5 GHz frequency band, which offers significantly more non-overlapping channels than the older 2.4 GHz band. This reduces co-channel interference and provides the wider bandwidth necessary for rigorous academic research, software downloads, and multimedia streaming. The network controller will implement active band-steering to force capable devices onto the 5 GHz spectrum, leaving the 2.4 GHz band strictly for older legacy devices. Additionally, seamless roaming protocols

802.11r(k,v) must be enabled so that student devices can transition between APs smoothly without dropping connections as they walk across campus.

For security, the network must implement WPA3-Enterprise authentication combined with an 802.1X RADIUS server. This ensures that every student logs in using their unique university credentials, allowing administrators to monitor individual usage, enforce bandwidth limits per user, and securely segment student traffic from faculty and administrative systems through dynamic VLAN assignment. A separate Captive Portal can be utilized for guest access, strictly isolating visitor traffic from internal academic resources.

This design perfectly balances the university's constraints. By using predictive RF heatmapping software before installation, the university can eliminate wireless dead zones and avoid over-purchasing hardware, rigorously adhering to the budget. Wi-Fi 6 and band-steering maximize the performance and lifespan of the limited APs. Finally, WPA3-Enterprise provides robust, enterprise-grade security with minimal ongoing operational costs, integrating flawlessly with the university's existing centralized student directory.